Interpretable Machine Learning

Additive Decomposition

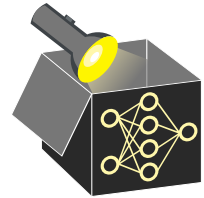
Learning goals

- What are additive decomposition of prediction functions?
- Why are they useful?
- How do we obtain them?

FUNCTIONAL DECOMPOSITION

▸ Li and Rabitz (2011)

▸ Chastaing et al. (2012)



For interpretation purposes, one might be interested in decomposing a square-integrable function $\hat{f} : \mathbb{R}^p \mapsto \mathbb{R}$ into sum of components of different dimensions w.r.t. inputs $x_1, \ldots, x_p$:

$$\hat{f}(\mathbf{x}) = \sum_{S \subseteq \{1,\ldots,p\}} g_S(\mathbf{x}_S) = g_\emptyset + g_1(x_1) + g_2(x_2) + \cdots + g_p(x_p) + g_{1,2}(x_1, x_2) + \cdots + g_{p-1,p}(x_{p-1}, x_p) + \cdots + g_{1,\ldots,p}(x_1, \ldots, x_p)$$

where

- $g_\emptyset \hat{=}$ Constant mean (intercept)
- $g_j \hat{=}$ first-order or main effect of j-th feature alone on $\hat{f}(\mathbf{x})$
- $g_S(\mathbf{x}_S) \hat{=}$ $|S|$-order effect, depends **only** on features in S

N.B.: A unique solution for the components only exists under certain assumptions

FUNCTIONAL DECOMPOSITION – ASSUMPTIONS

For independent inputs, the *vanishing condition* is required to obtain a unique solution:

$$\mathbb{E}_{X_j}(g_S(\mathbf{x}_S)) = \int g_S(\mathbf{x}_S) d\mathbb{P}(x_j) = 0, \forall j \in S, \forall S \subseteq \{1, \ldots, p\}$$

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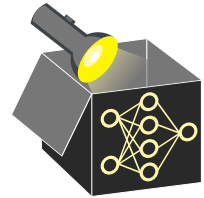

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Vanishing condition has the following implications:

- Marginalizing out $x_j, \forall j \in S$ for component $g_S(\mathbf{x}_S)$ yields a constant 0
  $\rightsquigarrow$ Makes sure that component $g_S(\mathbf{x}_S)$ does not contain effects of $x_j, \forall j \in S$
- Components are orthogonal (i.e., mutually independent and uncorrelated):

$$\mathbb{E}_X(g_V(\mathbf{x}_V) g_S(\mathbf{x}_S)) = 0, \forall V \neq S$$

- Variance can be decomposed: $Var[\hat{f}(\mathbf{x})] = \sum_{S \subseteq \{1,\ldots,p\}} Var\left[g_S(\mathbf{x}_S)\right]$

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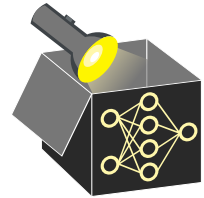

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N.B.: For dependent inputs, ▸ Hooker (2007) showed the existence of a unique solution for the components under a "relaxed vanishing condition" which leads to a "hierarchical orthogonality"

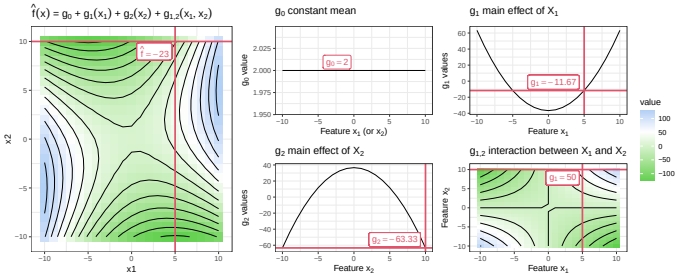
$$\mathbb{E}_X(g_V(\mathbf{x}_V) g_S(\mathbf{x}_S)) = 0, \forall V \subset S$$

$\rightsquigarrow$ Only components are orthogonal where features involved in $g_V(\mathbf{x}_V)$ also appear in $g_S(\mathbf{x}_S)$

FUNCTIONAL DECOMPOSITION – EXAMPLE

Example: $\hat{f}(\mathbf{x}) = 2 + x_1^2 - x_2^2 + x_1 \cdot x_2$ (e.g., if $x_1 = 5$ and $x_2 = 10 \Rightarrow \hat{f}(\mathbf{x}) = -23$)

- Computation of components using feature values $x_1 = x_2 = (-10, -9, \ldots, 10)^\top$ gives:



For $x_1 = 5$ and $x_2 = 10$:

- $g_\emptyset = 2$
- $g_1(x_1) = -9.67$
- $g_2(x_2) = -65.33$
- $g_{1,2}(x_1, x_2) = 50$

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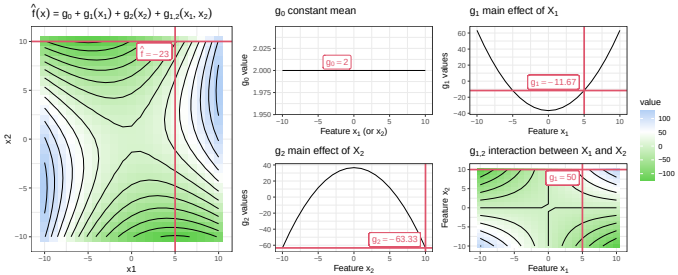

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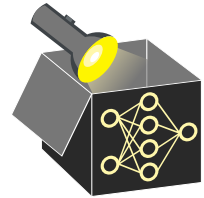
- Vanishing condition means:
  - g_1 and g_2 are mean-centered w.r.t. marginal distribution of x_1 and x_2
  - Integral of $g_{1,2}$ over marginal distribution x_1 (or x_2) is 0

FUNCTIONAL DECOMPOSITION – COMPUTATION



Computation of components via recursive expectations (where $-S = \{1, \ldots, p\} \setminus S$):

$$g_S(\mathbf{x}_S) = \mathbb{E}_{X_{-S}}\left[\hat{f}(\mathbf{x}) \mid x_S\right] - \sum_{V \subset S} g_V(x_V)$$

- Expectation integrates $\hat{f}(\mathbf{x})$ over all input features except $\mathbf{x}_S$
- Subtract all components g_V with $V \subset S$ to remove all lower-order effects and center the effect

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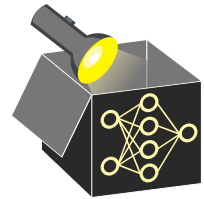

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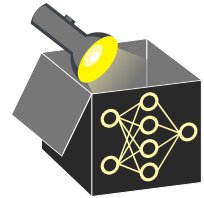
- Expectation integrates $\hat{f}(\mathbf{x})$ over all input features except $\mathbf{x}_S$
- Subtract all components g_V with $V \subset S$ to remove all lower-order effects and center the effect
- Recursive computation:

$$\begin{aligned}
g_\emptyset &= \mathbb{E}_X\left[\hat{f}(\mathbf{x})\right] \\
g_j(x_j) &= \mathbb{E}_{X_{-j}}\left[\hat{f}(\mathbf{x}) \mid x_j\right] - g_\emptyset, \ \forall j \in \{1, \ldots, p\} \\
g_{j,k}(x_j, x_k) &= \mathbb{E}_{X_{-\{j,k\}}}\left[\hat{f}(\mathbf{x}) \mid x_j, x_k\right] - g_k(x_k) - g_j(x_j) - g_\emptyset, \ \forall j < k \\
&\vdots \\
g_{1,\ldots,p}(\mathbf{x}) &= \hat{f}(\mathbf{x}) - \textstyle\sum_{S \subseteq \{1,\ldots,p-1\}} g_S(\mathbf{x}_S)
\end{aligned}$$

VARIANCE DECOMPOSITION

- Decomposition of $\hat{f}(\mathbf{x})$ allows to conduct functional analysis of variance (fANOVA)

VARIANCE DECOMPOSITION



- Decomposition of $\hat{f}(\mathbf{x})$ allows to conduct functional analysis of variance (fANOVA)
- If features are independent, variance can be additively decomposed without covariances:

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Var\left[\hat{f}(\mathbf{x})\right] &= Var\left[g_{\emptyset} + g_1(x_1) + \ \ldots \ + g_{1,2}(x_1, x_2) + \ \ldots \ + g_{1,\ldots,p}(\mathbf{x})\right] \\
&= Var\left[g_{\emptyset}\right] + Var\left[g_1(x_1)\right] + \ \ldots \ + Var\left[g_{1,2}(x_1, x_2)\right] + \ \ldots \ + Var\left[g_{1,\ldots,p}(\mathbf{x})\right]
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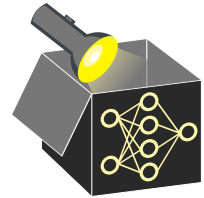

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\end{aligned}
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- Dividing by the prediction variance, yields fraction of variance explained by each term:

$$
1 = \frac{Var\left[g_{\emptyset}\right]}{Var\left[\hat{f}(\mathbf{x})\right]} + \frac{Var\left[g_1(x_1)\right]}{Var\left[\hat{f}(\mathbf{x})\right]} + \ \ldots \ + \frac{Var\left[g_{1,2}(x_1, x_2)\right]}{Var\left[\hat{f}(\mathbf{x})\right]} + \ \ldots \ + \frac{Var\left[g_{1,\ldots,p}(\mathbf{x})\right]}{Var\left[\hat{f}(\mathbf{x})\right]}
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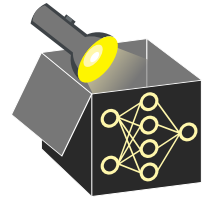

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- Fraction of variance explained by a component $g_V(\mathbf{x}_V)$ is the Sobol index:
  $S_V = \frac{Var[g_V(\mathbf{x}_V)]}{Var\left[\hat{f}(\mathbf{x})\right]}$
  $\rightsquigarrow$ Importance measure of component $g_V(\mathbf{x}_V)$
  $\rightsquigarrow$ Small S_V values $\Rightarrow$ Component g_V does not explain much of total variance of $\hat{f}$